PERIODIC TABLE BY SUBLEVEL

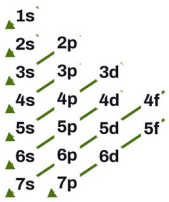
UNIT 2 TEST · REFERENCE · YOU MAY WRITE ON THIS SHEET ONLY IF I SAY SO

	s-BLOCK		d-BLOCK										p-BLOCK					
group	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
valence e⁻	1	2	valence = the electrons in the outer s sublevel										3	4	5	6	7	8
1s	1 H																	2 He
2s	3 Li	4 Be											5 B	6 C	7 N	8 O	9 F	10 Ne
3s	11 Na	12 Mg											13 Al	14 Si	15 P	16 S	17 Cl	18 Ar
4s	19 K	20 Ca	21 Sc	22 Ti	23 V	24 Cr	25 Mn	26 Fe	27 Co	28 Ni	29 Cu	30 Zn	31 Ga	32 Ge	33 As	34 Se	35 Br	36 Kr
5s	37 Rb	38 Sr	39 Y	40 Zr	41 Nb	42 Mo	43 Tc	44 Ru	45 Rh	46 Pd	47 Ag	48 Cd	49 In	50 Sn	51 Sb	52 Te	53 I	54 Xe
6s	55 Cs	56 Ba	71 Lu	72 Hf	73 Ta	74 W	75 Re	76 Os	77 Ir	78 Pt	79 Au	80 Hg	81 Tl	82 Pb	83 Bi	84 Po	85 At	86 Rn
7s	87 Fr	88 Ra	103 Lr	104 Rf	105 Db	106 Sg	107 Bh	108 Hs	109 Mt	110 Ds	111 Rg	112 Cn	113 Nh	114 Fl	115 Mc	116 Lv	117 Ts	118 Og

57–70 4f ↓ (between 6s and 5d)

89–102 5f ↓ (between 7s and 6d)

DIAGONAL RULE



FILLING ORDER, ROW BY ROW

Row 1	1s
Row 2	2s 2p
Row 3	3s 3p
Row 4	4s 3d 4p
Row 5	5s 4d 5p
Row 6	6s 4f 5d 6p
Row 7	7s 5f 6d 7p

SUBLEVEL CAPACITY

	ORBITALS	MAX e⁻
s	1	2
p	3	6
d	5	10
f	7	14

f-BLOCK · fits between groups 2 and 3 in rows 6 and 7

LANTHANIDES (4f)	57 La	58 Ce	59 Pr	60 Nd	61 Pm	62 Sm	63 Eu	64 Gd	65 Tb	66 Dy	67 Ho	68 Er	69 Tm	70 Yb
ACTINIDES (5f)	89 Ac	90 Th	91 Pa	92 U	93 Np	94 Pu	95 Am	96 Cm	97 Bk	98 Cf	99 Es	100 Fm	101 Md	102 No

READING A ROW

Read left to right and say the sublevel labels as you pass them. Row 4: 4s, then 3d, then 4p. The d sublevel is always one level lower than the row; f is two lower.

NOBLE-GAS SHORTHAND

Start from the noble gas at the end of the row above, in brackets: [He] [Ne] [Ar] [Kr] [Xe] [Rn]. Then write the rest of the row in filling order.